Ge Yan

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PhD candidate, Electrical Engineering, National University of Singapore, Singapore

RESEARCH INTERESTS

Reconfigurable wireless communication systems; channel knowledge map; wireless sensing; optimization algorithms; electromagnetic theory; computational imaging; machine learning.

EDUCATION

- **2022/08-2026.10 (expected)** **PhD. in Electrical and Computer Engineering**
  - Institution: National University of Singapore, Singapore
  - Advisors: Prof. Rui Zhang, Prof. Xudong Chen
  - Thesis Title: Reconfigurable MIMO for Wireless Communication: Optimization and Practical Channel Knowledge Acquisition
- **2021/01-2022/06** **MSc. in Electrical and Computer Engineering**
  - Institution: National University of Singapore, Singapore
  - GPA: 4.75/5.00
- **2016/09-2020/06** **B.Eng. in Information Science & Engineering**
  - Institution: Chien-Shiung Wu College, Southeast University, Nanjing, China
  - GPA: 3.89/4.00 (Overall Ranking: 1/114)

RESEARCH EXPERIENCE

- [Ongoing] **Channel Knowledge Map (CKM) Modelling from An Electromagnetic Perspective** (*2026/04-present*)
  - Develop an electromagnetic approach to CKM modeling and reconstruction in contrast to conventional ray-tracing-based approaches.
  - Represent wireless channels in a physically consistent manner without requiring detailed geometric descriptions of the environment.
  - Integrate physics-informed machine learning techniques for efficient CKM representation and the inverse problem, i.e., environment inference given the CKM.
- [Ongoing] **Slow Movable Antenna (MA)-Aided Wireless Communications Based on Long-Term Channel Statistics** (*2024/04-present*)
  - Developed general statistical channel models for MA-aided wireless systems across different timescales, establishing a novel system design paradigm based on long-term channel statistics rather than instantaneous channel state information (CSI).
  - Characterized long-term system performance with slow MAs and developed low-complexity antenna-position optimization algorithms requiring only minimal channel statistics.
  - Reduced the need for frequent channel acquisition and antenna repositioning, addressing key practical challenges in deploying MA-aided wireless systems.
  - Resulted in two first-authored journal papers (one under review), two conference papers, and one IEEE magazine paper in preparation.

- **Practical Configuration for Intelligent Reflecting Surface (IRS)-Assisted Wireless Systems** (*2022/10-2025/09*)
    - Developed low-complexity channel acquisition approaches for practical IRS-aided wireless systems by leveraging channel measurements available through existing protocols, which avoids additional signaling and facilitates seamless IRS integration into existing networks.
    - Proposed quasi-static IRS optimization algorithms for long-term coverage enhancement, reducing the need for frequent reconfiguration while improving communication performance cost-efficiently.
    - Resulted in three first-authored journal publications, one conference paper, and one IEEE magazine publication.

SELECTED PROJECTS & PROTOTYPING

At Prof. Rui Zhang's Research Group, National University of Singapore, Singapore.

- **Movable Antenna (MA)-Aided Wireless Communications Prototype** (*2022/10-2025/09*)
    - Developed a prototype with mechanical antenna positioning controls and integrated it with National Instruments software-defined radio platforms.
    - Designed and conducted experiments to validate the performance of MA-aided wireless communication algorithms under practical system conditions.
    - Led algorithm integration, system debugging, and performance evaluation.
- **Intelligent Reflecting Surface (IRS) Channel Acquisition Prototype** (*2022/10-2025/09*)
    - Collaborated with a remote implementation team to integrate algorithms from my research into a prototype of IRS-aided wireless communication systems.
    - Provided algorithm implementation and technical support, assisted with system debugging, and contributed to performance evaluation and validation.

PUBLICATIONS

- **Journals**
    1. **G. Yan**, L. Zhu, W. Ma, and R. Zhang, "Slow Movable Antenna System Design Based on Cell-Specific Long-Term Angular Power Spectrum," 2026. [Online]. Available: https://arxiv.org/abs/2605.12192. (submitted to *IEEE Trans. Wireless Commun.*)
    2. **G. Yan**, L. Zhu, and R. Zhang, "Movable Antenna Aided Multiuser Communications: Antenna Position Optimization Based on Statistical Channel Information," in *IEEE Trans. Commun.*, vol. 74, pp. 1793-1810, 2026.
    3. **G. Yan**, L. Zhu, and R. Zhang, "Power Measurement Enabled Channel Autocorrelation Matrix Estimation for IRS-Assisted Wireless Communication," in *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1832-1848, Mar. 2025.
    4. **G. Yan**, L. Zhu, H. Sun, and R. Zhang, "Wideband Coverage Enhancement for IRS-Aided Wireless Networks Based on Power Measurement," in *IEEE Wireless Commun. Lett.*, vol. 15, pp. 670-674, 2026.
    5. **G. Yan**, L. Zhu, and R. Zhang, "Passive Reflection Optimization for IRS-Aided Multicast Beamforming with Discrete Phase Shifts," in *IEEE Wireless Commun. Lett.*, vol. 12, no. 8, pp. 1424-1428, Aug. 2023.

6. L. Zhu, H. Mao, **G. Yan**, W. Ma, Z. Xiao, and R. Zhang, "Movable and reconfigurable antennas for 6G: Unlocking electromagnetic-domain design and optimization," *npj Wireless Technol.*, vol. 2, no. 1, pp. 1–10, Feb. 2026.

- **Magazines**
  1. H. Sun, L. Zhu, **G. Yan**, and R. Zhang, "Quasi-Static IRS Reflection Design for Network Coverage Enhancement: A Practical RSRP-Based Approach," in *IEEE Wirel. Commun.*, May 2026, early access, DOI: 10.1109/MWC.2026.3692436.
- **Conferences**
  1. **G. Yan**, L. Zhu, W. Ma, S. Tan, Q. Dong, and R. Zhang, " Slow Antenna Movement Design for Movable Antenna-Aided Wireless Systems Based on Cell-Specific Long-Term Angular Power Spectrum," (accepted for *IEEE Globecom 2026*).
  2. **G. Yan** and R. Zhang, " Antenna Position Optimization Based on Statistical Channel Information for Movable Antenna Aided Multiuser Communications," *2025 IEEE Globecom Workshops (GC Wkshps), Taipei, Taiwan*, 2025, pp. 514-519.
  3. **G. Yan**, L. Zhu, and R. Zhang, "Channel Autocorrelation Estimation for IRS-Aided Wireless Communications Based on Power Measurements," *2023 IEEE Globecom Workshops (GC Wkshps), Kuala Lumpur, Malaysia*, 2023, pp. 1457-1462.

PROPOSAL DEVELOPMENT EXPERIENCE

At Prof. Rui Zhang's Research Group, National University of Singapore, Singapore.

- Contributed to research grant proposals on MA-aided systems and channel knowledge maps, including system modeling, technical formulation, and experimental designs.

AWARDS & FELLOWSHIPS

- President's Graduate Fellowship (*2022/08-2026/07*) National University of Singapore
- National Scholarship (Top 1%, *2018/10*) Chinese Ministry of Education
- President Scholarship (Top 1%, *2017/10*) Southeast University, Nanjing, China

TEACHING EXPERIENCE

Served as a Teaching Assistant for multiple undergraduate and graduate courses throughout my PhD, all at the National University of Singapore; responsibilities included lecture material preparation, student consultation, assignment/quiz grading, tutorial instruction, and project supervision; skilled in identifying students' misconceptions and providing targeted guidance to facilitate conceptual understanding.

- EE4032 Blockchain Engineering (*2022 Fall, 2023 Fall, 2024 Fall*)
- CEG5103 Wireless and Sensor Networks for IoT (*2023 Spring, 2024 Spring*)
- EE5139 Information Theory and Its Applications (*2023 Fall*)
- CEG5102 Wireless Communications for IoT (*2025 Fall*)
- EE3131C Communication Systems (*2026 Spring*)
- EE5131/6131 Wireless Communications (Advanced) (*2026 Spring*)

PROFESSIONAL SERVICES

- Chaired session 3 of workshop 34 at IEEE Globecom 2025, Taipei, Taiwan.

- Mon, December 8, 2025, 14:00-15:30 in Room 101B at the Taipei International Convention Center (TICC).
- IEEE Graduate Student Member.
- Reviewer for IEEE journals (TWC, TCOMM, WCL, etc) and conferences (Globecom, ICC, VTC, etc).

INDUSTRY EXPERIENCE

- **2020/08-2020/12** **AI Research Intern, R&D, SenseTime, Shanghai, China**
  - Participated in image recognition and segmentation projects.
  - Investigated GAN-based video generation.

TECHNICAL SKILLS

Skilled in algorithm development and simulation conduction integrated with various physically grounded setups and efficient optimization tools, particularly for signal processing, estimation and optimization problems.

- **Wireless:** MIMO, beamforming, channel models, computational electromagnetics.
- **Algorithms:** optimization, numerical methods, signal processing, inverse problems.
- **Machine learning:** deep learning, computer vision, PyTorch, TensorFlow.
- **Programming & tools:** Python, MATLAB, C/C++, COMSOL, CST, CVX, Git, Linux.